World Health Organization
REGIONAL OFFICE FOR Europe

AirQ+: burden of disease due to air pollution manual

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December 2020

Abstract

AirQ+ is a software tool for quantifying the health burden and impact of air pollution developed by the WHO Regional Office for Europe. AirQ+ includes methodologies to assess the impacts of short- and long-term exposure to ambient air pollution. The main methodologies use evidence generated by epidemiological cohort studies showing a relationship between average long-term air pollution concentration levels and the mortality risks in exposed populations. Assessing the impact of air pollution is suggested when evaluating the consequences of policies and interventions or of hypothetical scenarios. AirQ+ should always be used with the support of an epidemiologist or air pollution impact assessment expert. To facilitate users in their analyses, AirQ+ comes with manuals that require increasing levels of expertise. This one shows how to carry out a burden of disease assessment for particulate matter with a diameter of 2.5 μm or less ($PM_{2.5}$) ambient air pollution. This is calculated by combining estimates of exposure to air pollution and its distribution in the population with results from epidemiological studies that indicate the additional disease burden from different levels of exposure to air pollution for specific causes of death. This calculation is particularly useful for global comparisons. For locations with high $PM_{2.5}$ ambient air concentration levels, it is recommended to use the integrated exposure–response functions (used for the Global Burden of Disease study) that cover cause-specific mortality for five causes of death: acute lower respiratory infections, chronic obstructive pulmonary disease, lung cancer, ischaemic heart disease and stroke.

Keywords

AIR POLLUTANTS
AIR POLLUTION – exposure
AIR POLLUTION – health impacts
AIR POLLUTION – burden of disease

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Introduction

According to WHO (2016),[1] the burden of disease of ambient air pollution is

estimated using methodology … based on combining estimates of exposure to air pollution and its distribution in the population, with results from epidemiological studies that indicate the additional disease burden from different levels of exposure to air pollution, after adjusting for other risk factors (e.g. tobacco smoke).

The burden of disease calculation for air pollution has generated various estimates, and the user should be aware of the changes in methodology that have occurred over the years (Ostro et al., 2018).[2] For locations where high $PM_{2.5}$ ambient air concentrations[3] (say, exceeding 40 µg/m^3) are prevalent, it is recommended to use the integrated exposure–response (IER) functions of the Global Burden of Disease (GBD) study[4] to quantify the mortality burden of air pollution. These risk functions cover cause-specific mortality for five specific causes of death: acute lower respiratory infections (ALRI), chronic obstructive pulmonary disease (COPD), lung cancer (LC), ischaemic heart disease (IHD) and stroke. Other health outcomes are associated with long-term exposure to particulate matter, such as adverse birth outcomes, childhood respiratory disease, diabetes and neurodegenerative disorders, but they are not considered in this version of AirQ+.

This manual shows how to carry out a burden of disease[5] assessment for $PM_{2.5}$ ambient air pollution (a well-established risk factor), using the default GBD IER functions included in the current version of the AirQ+ software. It is worth noting that the GBD Risk Factor Collaborators update the IER functions periodically.[6] Consequently, reproducing and comparing with GBD results is difficult given the changes in methodology and the modelling used to produce GBD input data (Ostro et al., 2018).

AirQ+ can also be used to estimate the burden of household air pollution related to solid fuel use (SFU)[7] using a simplified method that provides a rough estimate of the implied burden of indoor air pollution.

1 WHO (2016). Ambient air pollution: a global assessment of exposure and burden of disease. Geneva: World Health Organization (https://www.who.int/phe/publications/air-pollution-global-assessment/en/, accessed 25 November 2019).

2 Ostro B, Spadaro JV, Gumy S, Mudu P, Awe Y, Forastiere F, et al. (2018). Assessing the recent estimates of the global burden of disease for ambient air pollution: methodological changes and implications for low- and middle-income countries. Environ Res. 166: 713–25. doi:10.1016/j.envres.2018.03.001.

3 $PM_{2.5}$: particulate matter with a diameter of 2.5 µm or less

4 GBD 2017 Risk Factor Collaborators (2018). Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. Lancet 392:1923–94 (https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(18)32225-6/fulltext).

5 Burden of disease estimates are important because they highlight the adverse health effects of air pollutions, put a spotlight on the importance of environmental protection measures that aim to safeguard public health, and support evidence-based policymaking. This type of information is also often used for cross-country comparisons, assuming age-standardized population distributions, with the underlying scope to show the avoidable health losses that could be achieved from improvements in air quality due to control measures that reduce pollutant emissions.

6 Further details are available at: IHME (2019). Global Health Data Exchange [website]. Seattle (WA): Institute for Health Metrics and Evaluation (http://www.healthdata.org/, accessed 25 November 2019).

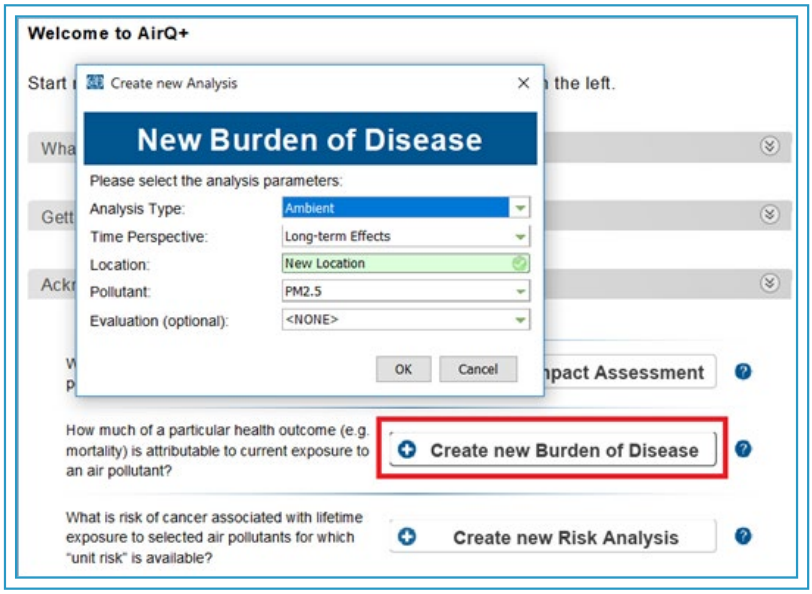
7 For details on key terms please refer to the AirQ+ Glossary, which is accessible on the main welcome window.

It is recommended to always use AirQ+ with the support of an epidemiologist or air pollution impact assessment expert.

Create new burden of disease analysis

From the main AirQ+ welcome window, click the "Create new Burden of Disease" button to get to the "New Burden of Disease" window (Fig.1).

Fig. 1. New Burden of Disease window



The user selects options from the drop-down boxes to indicate the type of analysis to perform and clicks OK. This brings up the Analysis Properties window, where the user can enter data for the analysis. After entering data, the user moves to the New Burden Evaluation window where the Health Endpoint drop-down box lists all specific causes of mortality for which IER functions are available. In the current version of AirQ+, the user may select from IER 2013, 2015 and 2016. Each of these models has an implied cut-off value, and the lowest level at which health effects have been found is indicated in the Burden Evaluation window. The cut-off value cannot be modified by the user (refer to the Glossary for details on the methodology). The user should check the age groups to which the IER function refers. IER functions are age-specific to the cause of death. For example, both IHD and stroke analyses can be performed on adults by age group (in years): 25–29, 30–34, 35–39 ... 95+. The most recent IER functions support analyses on ALRI in children (0–5 years old), COPD in adults (>25 years old) and lung cancer in adults (>25 years old).

The burden of disease attributable to ambient air pollution is estimated by combining exposure to air pollution and its distribution in the population with IER estimates at each level of exposure. The attributable burden is calculated by multiplying the population-attributable fraction by the health outcome for each health outcome, sex and age group (WHO, 2016). The total number of deaths for the selected study area has to be estimated using the IER functions for each health outcome: ALRI, lung cancer, COPD, IHD and stroke. The next example discusses the calculations for one health outcome, lung cancer.

Example CityData analysis:

ambient air pollution – $PM_{2.5}$ – long-term – adult mortality – use of IER

Question to be addressed: What is the health burden from lung cancer in adults at the current level of $PM_{2.5}$ air pollution?

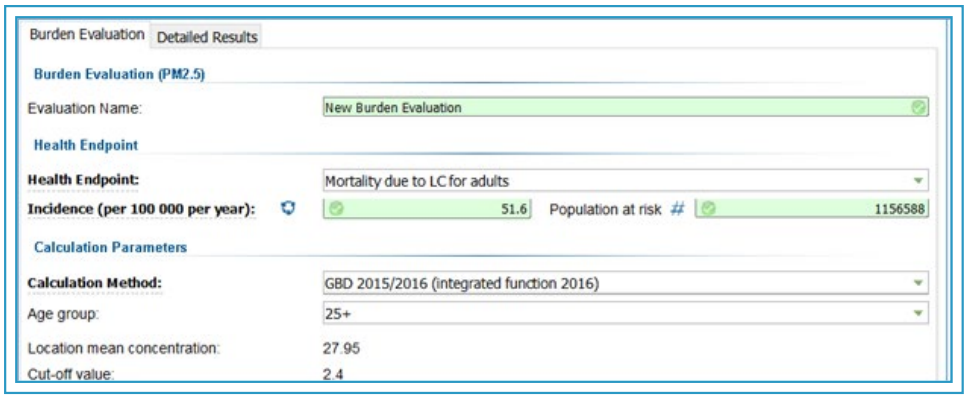
Data input: $PM_{2.5}$ yearly average

In the "New Burden of Disease" window (Fig.1), select long-term effects, ambient pollution and the pollutant $PM_{2.5}$. In the Analysis Properties window, select the Input Mean Value option and enter 27.95 µg/m^3 into the Mean Value field as the value for ambient air $PM_{2.5}$ concentration. Enter "1156588" into the Total Population field. AirQ+ does not use the total population for the analysis, so the user can consider this as additional descriptive information.[8]

In the Burden Evaluation window, select "Mortality due to LC for adults" in the Health Endpoint drop-down box and enter:

- mortality incidence for adults, LC, per 100 000 population: 51.6
- the default percentage of the total number of adults exposed to the pollutant: 100%.

Fig. 2. AirQ+ Burden Evaluation window for ambient air pollution – $PM_{2.5}$ – long-term – adult LC mortality – use of IER

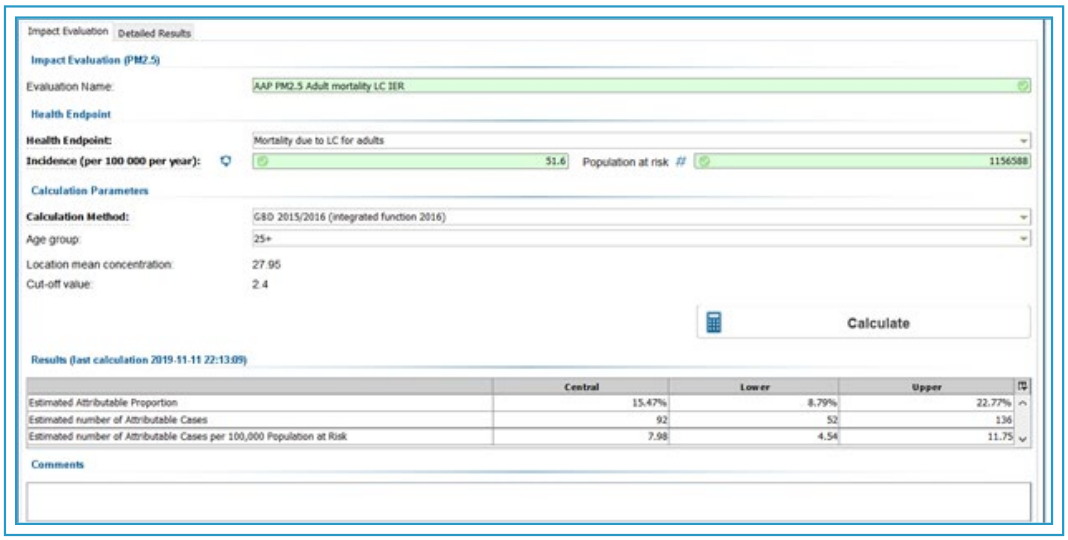


8 AirQ+ uses the total number of adults (≥ 30 years) exposed to the pollutant, a value the user enters when creating a new Burden Analysis.

Results

The percentages for the attributable proportion of air pollution on lung cancer are high, due to the high hypothesized incidence, and this produces the estimate of 92 deaths (central estimate) (Fig. 3). If the user has other cause-specific mortality data available, the results can be summed up and the whole mortality due to ambient air pollution can be estimated.

Fig. 3. AirQ+ results for air pollution – $PM_{2.5}$ – long-term –use of IER LC mortality (mean concentration: 27.95 µg/m^3)



Example CityData analysis:

household air pollution – SFU – long-term – child mortality due to ALRI

The household air pollution module in AirQ+ provides a rough estimate of the health burden from exposure to indoor air pollution. It is also meant to promote the collection of data on household air pollution.

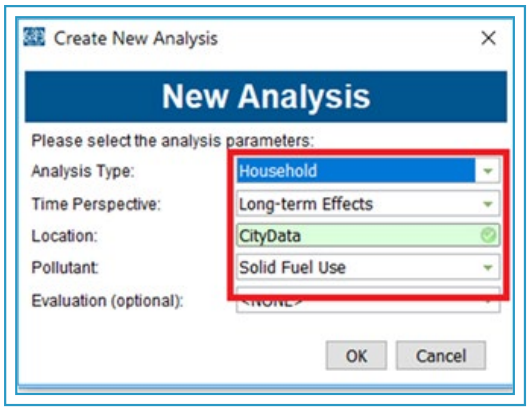
Question to be addressed: How many children are affected by ALRI because of the current level of household (indoor) air pollution due to solid fuel use?

Data input

In the New Analysis window, select household, long-term and SFU as the pollutant to analyse (Fig. 4). In the Analysis Properties window, enter "105352" for the population of interest (children under 5 years. In the Burden Evaluation window, enter the following data:

- estimate of population (children) exposed to the pollutant (SFU) as either the percentage (11%) of or total number ("11589") of children;
- mortality incidence for children, ALRI per 100 000 population: 5000; and
- the default relative risk values for mortality due to ALRI (GBD 2012): 2.9 (95% CI 2.0–3.8).

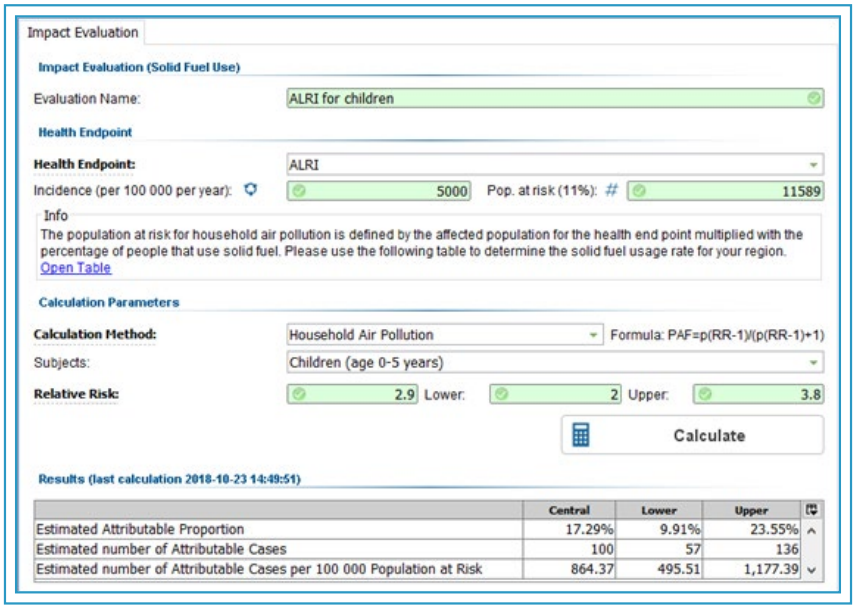
Fig. 4. AirQ+ New Analysis window for household air pollution – SFU – long-term – child mortality due to ALRI (data input – SFU)



Results

Fig. 5 summarizes the results of this analysis. The $PM_{2.5}$ attributable mortality due to ALRI is just over 17% (95% CI: 10–24%), with 100 (95% CI: 57–136) deaths among children younger than 5 years in this hypothetical case study.

Fig. 5. AirQ+ results for household air pollution – SFU – long-term – child mortality due to ALRI



The WHO Regional Office for Europe

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